When iterated $n+1$ times, we can see the scheme $[\![\text{OABA}]\!]$ produces a mapping of the form

$$\left(\mathscr{U}_h^{[\![\text{OABA}]\!]}\right)^{n+1} = \mathscr{U}_{h/2}^{\text{A}} \circ \mathscr{U}_{h/2}^{\text{B}} \circ \left(\mathscr{U}_h^{[\![\text{BAOAB}]\!]}\right)^{n} \circ \mathscr{U}_{h/2}^{\text{B}} \circ \mathscr{U}_{h/2}^{\text{A}} \circ \mathscr{U}_h^{\text{O}},$$

therefore through a coordinate transformation the $[\![\text{OABA}]\!]$ method can be morphed into the $[\![\text{BAOAB}]\!]$ scheme. To put this another way, if we compute an iteration using $[\![\text{OABA}]\!]$ and simply evaluate it at different points within the step, we can recover the same results as the $[\![\text{BAOAB}]\!]$ method would provide. Another term to describe this relationship is to say that one method is *conjugate* to the other.

7.3.2 Other Langevin Integrators

We may use our alphabet of A, B and O splitting pieces to create a wide variety of methods other than those given in the previous section (some requiring multiple force evaluations). However, it is not obvious that there is any gain in increasing the complexity of the splitting scheme (in terms of the length of its codifying string). Some popular methods can be rewritten in terms of this splitting, for example the scheme considered in [62] is equivalent to $[\![\text{OBABO}]\!]$. Examples of methods based on this type of splitting were also proposed by De Fabritiis et al [96] and Thalmann and Farago [363], in the context of dissipative particle dynamics (see Chapter 8).

Other Langevin dynamics splitting schemes in common use utilize a different additive decomposition of the SDE vector field. For example, the Stochastic Position Verlet (SPV) method of [265] relies on the splitting strategy

$$\mathrm{d}\begin{bmatrix}\boldsymbol{q}\\ \boldsymbol{p}\end{bmatrix} = \begin{bmatrix}\boldsymbol{M}^{-1}\boldsymbol{p}\\ \mathbf{0}\end{bmatrix}\mathrm{d}t + \begin{bmatrix}\mathbf{0}\\ -\nabla U(\boldsymbol{q})\mathrm{d}t - \gamma\boldsymbol{p}\mathrm{d}t + \sigma\boldsymbol{M}^{1/2}\mathrm{d}\mathbf{W}\end{bmatrix}.$$

To solve the splitting piece including the stochastic $\mathrm{d}\mathbf{W}$ term, we note that $\boldsymbol{q}$ does not change through its integration, and hence we may treat $\boldsymbol{F} = -\nabla U$ as constant during this part of the step. The exact solution is hence given through the usual Ornstein-Uhlenbeck formula, where the scheme itself derives from a symmetric integration of the two parts:

Stochastic Position Verlet (SPV)

$$\begin{aligned}\boldsymbol{q}_{n+1/2} &= \boldsymbol{q}_n + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p}_n\\ \boldsymbol{p}_{n+1} &= e^{-h\gamma}\boldsymbol{p}_n - \eta\nabla U(\boldsymbol{q}_n) + \zeta\boldsymbol{M}^{1/2}\mathbf{R}_n\\ \boldsymbol{q}_{n+1} &= \boldsymbol{q}_{n+1/2} + (h/2)\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1}\end{aligned}$$

where $\eta = (1-e^{-h\gamma})/\gamma$, $\zeta = \left[k_{\text{B}}\text{T}\left(1-e^{-2\gamma h}\right)\right]^{1/2}$. It is important to note that in the limit of infinite friction, we have $\eta \to 0$ independent of the timestep h. This suggests that this splitting strategy is unsuitable for large choices of the parameter γ as the

sampling will become inconsistent: the force will be dominated by the stochastic term, leaving the dynamics to evolve as a random walk.

Similar behavior is exhibited using the splitting strategy in (7.5), when interposing the *kick* term B between two O steps, such as in the schemes ⟦AOBOA⟧ or ⟦BOAOB⟧. By avoiding "OBO" updates, we can derive schemes that are consistent even at infinite friction (the ⟦BAOAB⟧ and ⟦ABOBA⟧ schemes are examples of such methods).

The Brünger-Brooks-Karplus (BBK) Langevin integrator [55] is one of the most popular and widely-implemented schemes.

Brünger-Brooks-Karplus (BBK)

$$\begin{aligned}
\boldsymbol{p}_{n+1/2} &= (1 - h\gamma/2)\boldsymbol{p}_n - (h/2)\nabla U(\boldsymbol{q}_n) + \tfrac{1}{2}\sqrt{2k_{\mathrm{B}}\mathrm{T}h\gamma}\boldsymbol{M}^{1/2}\mathbf{R}_n \\
\boldsymbol{q}_{n+1} &= \boldsymbol{q}_n + h\boldsymbol{M}^{-1}\boldsymbol{p}_{n+1/2} \\
\boldsymbol{p}_{n+1} &= [\boldsymbol{p}_{n+1/2} - (h/2)\nabla U(\boldsymbol{q}_{n+1}) + \tfrac{1}{2}\sqrt{2k_{\mathrm{B}}\mathrm{T}h\gamma}\boldsymbol{M}^{1/2}\mathbf{R}'_n]/(1 + h\gamma/2)
\end{aligned}$$

Often $\mathbf{R}'$ is chosen such that $\mathbf{R}'_n = \mathbf{R}_{n+1}$, so the random vector would be reused at the start of the following timestep, however there are many variants of this scheme that incorporate the redraw of random numbers differently. In other variants of BBK we take $\mathbf{R}'_n = \mathbf{R}_n$, reusing the random number within the step. We may also consider independent $\mathbf{R}'_n$ and $\mathbf{R}_n$, however when using independent noise processes we must modify the algorithm as it will not sample canonically (see Exercise 1).

7.4 Error Results for Harmonic Problems

For general systems it can be very challenging to give a complete description of the weak error in averages computed from trajectories evolved through a Langevin dynamics discretization scheme. However, using a linear model problem significantly reduces the complexity, while using small dimensional test problems allows us to effectively annihilate sampling error in demonstrative numerical experiments.

As such, we return now to the harmonic oscillator, which as well as being the simplest molecular model is also one of the most relevant for molecular dynamics applications, as many issues of stability and timestep in molecular dynamics simulations arise due to harmonic potentials used to model covalent bonds (such as in crystalline solids and biomolecules). The canonical distributions of position and momentum are also of a simple form (Gaussians), making such oscillators particularly amenable to analysis.

Recall that the one-dimensional harmonic oscillator with spring constant Ω^2 has potential energy function $U(q) = \Omega^2 q^2/2$, with Hamiltonian

$$H(q,p) = \frac{p^2}{2m} + \frac{\Omega^2 q^2}{2}, \tag{7.6}$$

where $(q,p) \in \mathbb{R}^2$ represent a particle's position and momentum, and $m > 0$ is the particle's mass. In this setting, the canonical distribution ρ_β is the product of two